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Unit – 5 ➡ Concepts & Awareness

Introduction:

- Pollution and its effects have severed ecosystems and the general balance of organisms.
- Due to anthropogenic activity “waste” is generated.
- It is another reason/source causing or accelerating pollution.
- Waste management therefore comes into the picture.
- Proper waste management could help reduce pollution and acquire sustainability.

Waste:

- Waste is defined as unwanted and unusable materials and is regarded as a substance that is of no use.
- Garbage is mainly considered a solid waste that includes wastes from our houses, wastes from schools and offices, and wastes from industries and factories.
- To deal with such issue, concepts of 4R's is useful.

Concept of 4R's

→ The concept of 4R's is principle referring to the four words starting with the alphabet R. They are:

1. Reduce:

- we must try to reduce the use of non-biodegradable compounds which are harmful to the whole nature and all the beings living in it.
- The non- biodegradable compounds such as plastics, do not degrade upon being thrown into the landfills as well, the plastics can stay non degrading up to 500 years.
- try to minimize the unnecessary use of artificial things or products in our daily life.
- choose durable items while buying.
- Use washable containers for lunches.
- Reduce use of non-reusable plastic bags.
- We should buy recycled product.
- It is better to use cotton bags for shopping.
- Print on both sides of a paper so that we can reduce waste production.
- Warning signs should be printed on products showing the disposal method.

2. Reuse:

- the non-biodegradable waste causes a lot of damage to earth, nature and all the living beings.
- One of the ways to reduce the non-biodegradable footprint is by reusing the plastics. Most plastic bags can be reused multiple times before finally becoming non usable.
- Hence, by reusing most of the non-biodegradable things we throw away, we can reduce the harmful waste being released into nature.
- The initial cost of reusable products is high but is highly beneficial from a long-term view.
- Use reusable cotton bags for shopping.
- We should donate our old clothes & other items to charitable organizations.
- Buy a rechargeable battery.

- Reuse electronics devices after repairing.

3. Recycle:

- This process is a continuation of the previous process of Reusing.
- The recycle process refers to the process where in the waste plastics, that can't be reused and will have to be thrown away goes through a process, wherein another product is made out of the waste.
- Through the recycle process, the non-biodegradable waste instead of going to any landfill as a waste, instead it becomes another useful product.
- Paper is recycled to cardboard, toilet paper, tissue paper, etc.
- The newspaper is recycled into egg cartons, shoe boxes, etc.

4. Recover:

- This final process is important to ensure that the maximum part of the waste that is harmful for nature, must be recovered through either the Reuse process or the Recycle process.
- During the above three processes, there is always a possibility of non-biodegradable waste being released into the nature, recovering back most of it and reusing and recycling is necessary for the nature to still be sustainable.
- Most of the materials thrown in the garbage can be used & processed in ways other than being destroyed. Such a process is recovering.
- Methane gas can be recovered from decomposing organic matters and can be used to make biogas for domestic purpose.
- If there is still plastic waste left over after being reduced, reused, and recycled.
- It can be used to make energy. It is the process to convert waste products into resources (such as fuel, heat, etc.) through biological/mechanical means.

Importance of 4Rs

- To utilize the waste materials.
- To maintain environmental health.
- To minimize land and water pollution.
- To reduce consumption of needless items.
- To promote the use of degradable products.
- To utilize products to their maximum life.

Comparison of 4Rs

- The order of preferability of 4Rs are
 1. Reduce
 2. Reuse
 3. Recycle
 4. Recover
- Producing or generating low amount of waste, helps minimizing the effort to manage and act upon the waste.
- Prevention is better than cure.
- Recycle and recover method uses extra external resource to convert waste into useful entity so they are less favourable.
- Least favourable option is disposal of the waste generated.
 - As it pollutes land or water resource.

NGO

- NGO refers to non-governmental organisation.
- A non-governmental organization (NGOs) is any non-profit, voluntary *citizens' groups* which is organized on a local, national or international level.
- India has a number of NGOs that work in the field of environmental conservation and ecology.
 - Clean India, TERI (Tata Energy Research Institute), Sulabh international, etc.
- They are simply agencies or groups which are different from government bodies, which is characterized by creativity in approach, flexibility in functioning and innovativeness in methods and technologies.
- In recent years role of NGO in increasing environmental awareness has increased.

Role of NGO

- NGO works towards analysis and monitoring of natural resources and spreads information and ways to conserve it and educate more mass.
 1. Creating awareness among the public on current environmental issues and solutions.
 2. Facilitating the participation of various categories of stakeholders in the discussion on environmental issues.

3. Being involved in the protection of human rights to have a clean environment.
4. Protecting the natural resources and entrusting the equitable use of resources.
5. Data generation on natural resources, time line history of villages.
6. Analysis and monitoring of environmental quality.
7. Transferring information through newsletters, brochures, articles, audio visuals, etc.
8. Organizing seminars, lectures and group discussion for promotion of environmental awareness.
9. Helping the villages' administrative officials in preparation, application and execution of projects on environmental protection.

Environmental ethics

- Environmental ethics refers to the issues, principles and guidelines relating to human interactions with their environment.
- Environmental ethics is the discipline that studies the moral relationship of human beings to, and also the value and moral status of, environment and its non-human contents.
- It is part of environmental philosophy which considers extending traditional boundaries of ethics from solely including humans to including the non-human world.
- It exerts influence on a large range of disciplines including law, sociology, theology, ecology, economics and geography.
- There are many ethical decisions that human beings make with respect to the environment, like
 - Should we continue to cut tree for the sake of human consumption?
 - Should we continue to make gasoline powered vehicles?

Objectives of environmental ethics

- Environmental ethics helps in guiding human to take up consideration of future and other non-human content with respect of development.
 1. The primary goal of environmental ethics is to preserve environment to its best value.
 2. Offer moral justification for social policies aimed at protecting the environment.
 3. Reversing environmental degradation.
 4. Preserve natural resource and ecological balance.

Ethical theories

→ There are 3 approaches to environmental ethics.

1. Anthropocentric/human centric:

- humans are the dominant and important species on the planet earth.
- That gives them the power to manipulate and use nature for their own benefit

2. Life centric:

- humans have the ethical responsibility towards future generations of human beings, who must leave earth in a good condition for future generations.

3. Eco centric:

- An extension of the above view seeks respect for all life and demands reverence towards the entire environment.
- Such a non-anthropocentric approach that talks of ethical responsibility towards other species and even ecosystems.

Environmental ethics in India

→ Environmental ethics, the peaceful coexistence with nature has direct or indirect mention in our constitution.

→ An article in DPSP in Indian constitution,

- Article 48-A: State shall endeavor to protect and improve environment and to safeguard forest and wildlife of the country.
- Article 51-A (g): It is the duty of every citizen of India to protect and improve natural environment including forests, lakes, rivers, wildlife, and to have compassion for living creatures.

→ India has passed several laws for wildlife and ecosystem conservation.

- The Wildlife (Protection) Act, 1972.
- The Water (Prevention and Control of Pollution) Act, 1974.
- The Air (prevention and control of pollution) act, 1981: To provide for prevention, control, and abatement of air pollution. To provide for the establishment of the boards at the central and state levels to implement the act.
- The Environment (Protection) Act, 1986: This act was passed under article 253. This was passed in the wake of the Bhopal gas tragedy in December 1984.